

Jenny Liu

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E-portfolio: https://jennyhzliu.github.io/E_Portfolio/hl2633@cornell.edu

EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Bachelor of Science, Intended Mechanical Engineering major

Aug 2026-Expected May 2030

PROFESSIONAL EXPERIENCE

Valiant TMS, Windsor, Canada, *Industrial Design Group Intern*

Jun 2024 - Jul 2024

- Created 5 preliminary 3D models from engineering drawings in Autodesk Inventor for existing projects
- Participated in and documented 3 supplier design meetings involving conveyor belts, shaft collars, and flywheels

LEADERSHIP EXPERIENCE

Robotics Club, Branksome Hall High School, *Club Head*

Sep 2024 - May 2026

- Organized and led meetings with 20+ robotics club members for 6 hours weekly across 3 days
- Guided teams through end-to-end robot development and participated in 5 VEX competitions annually
- Earned VEX Judges Award and Semifinalist recognition in RECF Community Challenge, presented by Google

World Affairs Conference, *Technology Department Head*

Sep 2024 - Mar 2026

- Maintained and enhanced an existing 1,500+ line web application using JavaScript, HTML, CSS; integrated with Firebase for student-teacher registration; coordinated development and issue tracking through GitHub Projects
- Supported technology operations for a conference serving 1,200+ participants from 15+ Toronto schools

RESEARCH EXPERIENCE

“SNAP-HAR: Signal-Neural Adaptive Processing for Robust Human Activity Recognition” **May 2025 - Jul 2025**

Independent Researcher - CONF-MLA 2025

- Developed a machine-learning framework for robust wearable-sensor activity recognition using VS Code
- Evaluated 3 datasets across 5 masking strategies, improving F1-score by 17.0% on clean and 24.6% on noisy data

“Analysis of the Driving Stability of a Multi-Model Drive Electric-Tracked Intelligent Detection Device”

Jilin University, *High School Research Intern* - Actuators, 13(12), 2024

Apr 2024 - Nov 2024

- Developed a virtual prototype simulation model using RecurDyn and MATLAB and conducted literature review through 25+ hours of research per week
- Collected and analyzed forward, vertical, and roll angular acceleration data from a physical prototype across three operating conditions to evaluate driving stability

PASSION PROJECTS

Emergency Door Release Mechanism in Electric Vehicle Fires, *Independent Researcher*

Nov 2023 - Mar 2024

- Designed and prototyped an emergency EV door-release mechanism in SolidWorks; conducted 65 trials across 13 test conditions to evaluate reliability and analyzed experimental data using Python
- Earned Gold Medal at the Toronto Science Fair and presented the project at a school-wide STEM assembly

Drinking Fountain Redesigned, Industrial Design, *Independent Researcher*

Nov 2022 - Mar 2023

- Used Inventor and Blender to improve accessibility (stretching tubes, sterilization lamps, water dispenser)

Ultralow friction of DLC films for space tribology applications, *Independent Researcher*

May 2023 - Aug 2023

- Conducted literature review regarding hydrogen DLC films' composition and performance in low-earth orbits

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, HTML/CSS

Mechanical Design: SolidWorks, Autodesk Inventor, 3ds Max, Blender

Simulation & Analysis: RecurDyn, MATLAB

Language: English (bilingual); Mandarin (bilingual); Spanish (introductory)